

Grace Xiao

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EDUCATION

University of California, San Diego

BS in Mechanical Engineering, 4.0 GPA

La Jolla, CA

Expected Graduation June 2028

Apple Next-Gen Innovators Mentorship Program

Apple Inc.

La Jolla, CA

Sept 2025 – Present

- Selected as 1 of 11 UCSD students for Apple's competitive hardware engineering mentorship program
- Engaged in 1-on-1 mentorship with Apple engineer, actively improving technical understanding (AI, robotics) and leadership skills through continuous feedback and application
- Participated in technical workshops and connected with engineers across different teams, gaining exposure to mechanical design and hardware development

EXPERIENCE

Undergraduate Researcher

Multi-Agent Robotics Lab

Sept 2025 – Present

La Jolla, CA

- Implemented robust Model Predictive Control (MPC) algorithms for multi-agent coordination and dynamic obstacle avoidance
- Simulated autonomous TurtleBot navigation using ROS2 Humble and Gazebo
- Conducted physical experiments on OptiTrack for validation and hardware integration

CAD Assistant

Nano Bioelectronics (NBE) Lab

April 2025 – Jan 2026

La Jolla, CA

- Designed and rendered CAD models of multi-layer flexible wearable biosensors using Fusion 360 and AutoCAD
- Modeled anatomical environments (e.g., skin layers, finger topology) for fabrication-ready CAD schematics

Instructional Assistant

MAE 8 Tutor

Aug. 2025 – Sept. 2025

La Jolla, CA

- Hosted 1:1 sessions and provided real-time debugging of student code to improve class performance in MATLAB programming
- Conducted daily technical lab sessions and coordinated teaching goals through weekly meetings with the instructional team

PROJECTS

Coursework Development

Jacobs School of Engineering

July 2025 – Present

La Jolla, CA

- Collaborated with instructors and engineering faculty to develop and refine educational content focused on problem solving and engineering concepts
- Winner of "Outstanding Engineering Projects" at the 2026 San Diego County Engineering Honor Awards

Lead Engineer - Liquid Bi-propellant Rocket Riptide

SEDS at UCSD

Dec. 2024 – Jan. 2026

La Jolla, CA

- Led the controls team in the design and implementation of the rocket control system, including hardware and software integration
- Developed inverse kinematics based Thrust Vector Control system using multithreaded Python/C++
- Modeled gimbaled engine in CAD and actuated via CAN bus
- Conducted computational fluid dynamics simulations in ANSYS Fluent and OpenFOAM to model air flow around rocket

Events and Outreach Officer

Envision Makerspace

Nov. 2024 – June 2025

La Jolla, CA

- Led hands-on workshops on 3D printing, laser cutting, and other fabrication tools with 20+ attendees
- Assisted students in learning and increased access to makerspace resources